

Notes: Aghion, Burgess, Redding, and Zilibotti (AER, 2008)
**The Unequal Effects of Liberalization: Evidence from Dismantling the License
Raj in India**

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1 One-sentence summary

A nationwide product-market deregulation in India raised manufacturing output mainly in states with pro-employer labor regulations, showing that the gains from liberalization depend strongly on local labor-market institutions.

2 Big picture

2.1 Research question

The paper asks whether a common national reform - dismantling the License Raj in Indian manufacturing - had different effects across states depending on labor-market institutions. The central question is not only whether deregulation works on average, but whether it works differently in pro-employer versus pro-worker environments.

2.2 Main empirical finding

Industries located in states with pro-employer labor regulations grew faster after delicensing than the same industries located in pro-worker states. The key coefficient is the interaction between delicensing and state labor regulation. It is negative because the labor regulation index is higher when regulation is more pro-worker; hence, delicensing has weaker output effects in pro-worker states.

2.3 Why the result matters

The result implies that product-market reforms and factor-market institutions are complements. Removing entry and capacity controls can expose firms to competition and new investment opportunities, but the ability to respond depends on whether firms can adjust employment and production efficiently.

2.4 Main takeaway

The paper shifts the focus from a simple reform-versus-no-reform comparison to a more institutional question: the same liberalization can magnify regional inequality when local institutions differ.

3 Incremental contribution of the paper

3.1 From average liberalization effects to heterogeneous effects

Earlier work often asked whether deregulation, trade liberalization, or industrial delicensing raised output on average. This paper's incremental contribution is to show that the average effect hides large heterogeneity across institutional environments.

3.2 Interaction between product-market and labor-market regulation

The paper provides an early empirical test of a core idea in reform economics: product-market deregulation may require complementary labor-market flexibility. Delicensing removes output and entry controls, but pro-worker labor laws may prevent firms from adjusting employment or reallocating resources.

3.3 Staggered nationwide reform with local institutions

The Indian setting is powerful because delicensing occurs at the industry-year level and applies nationwide. This allows the authors to compare the same delicensed industry across states with different labor regulations, using state-industry fixed effects and industry-year fixed effects.

3.4 Institutions as mediators of reform

The paper moves beyond treating institutions as direct determinants of growth. It shows that institutions mediate the effects of policy shocks: labor regulation becomes more consequential after the product-market constraint is relaxed.

3.5 Distributional implication of reform

The paper's result is not simply that reform raises efficiency. Liberalization can widen regional disparities by benefiting states with more flexible institutions. This is an important political-economy lesson for reform sequencing.

4 Data and measurement

4.1 Industrial performance data

The main industrial outcome data come from the Annual Survey of Industries. The sample is a balanced panel of three-digit state-industries from 1980 to 1997. The unit is a state $\times$ three-digit industry $\times$ year observation.

- The sample includes 16 major Indian states.
- It covers an average of 64 three-digit manufacturing industries.
- The panel contains 18,324 observations.
- Outcomes include real output, number of factories, employment, and fixed capital.

4.2 Delicensing measure

The delicensing variable equals one once all or part of a three-digit industry is exempted from industrial licensing. The paper codes delicensing dates from industrial policy statements, press notes, and government notifications.

4.3 Labor regulation measure

State labor regulation is based on state amendments to the Industrial Disputes Act. Amendments are coded as:

$$+1 = \text{pro-worker}, \quad 0 = \text{neutral}, \quad -1 = \text{pro-employer}.$$

The cumulative score over time creates the labor regulation index. A higher value means a more pro-worker environment.

4.4 Trade and FDI controls

Because the 1991 reforms also reduced tariffs and liberalized foreign direct investment, the paper constructs controls for:

- applied tariff rates by three-digit industry;
- the fraction of Harmonized System six-digit products within a three-digit industry opened to automatic FDI approval.

4.5 State and state-industry covariates

Robustness checks include interactions of delicensing with state development expenditure, predicted financial development, industry productivity terciles, and political-party controls. These checks ask whether labor regulation proxies for other state capabilities or political environments.

5 Empirical specification and interpretation

5.1 Average delicensing effects

A baseline specification estimates average effects of delicensing:

$$Y_{ist} = \alpha_{is} + \lambda_t + \beta \text{Delicense}_{it} + \varepsilon_{ist},$$

where i indexes industry, s state, and t year. The main outcomes are log number of factories and log real output.

Detailed interpretation: The average effects are modest. Delicensing significantly raises the number of factories, but the average output effect is small and insignificant. This motivates looking at heterogeneous effects across states.

5.2 Main interaction specification

The core regression is:

$$Y_{ist} = \alpha_{is} + \lambda_{st} + \mu_{it} + \beta(\text{Delicense}_{it} \times \text{LaborReg}_{st}) + \varepsilon_{ist}.$$

The fixed effects are demanding:

- state-industry fixed effects absorb permanent comparative advantage;
- state-year fixed effects absorb statewide shocks;
- industry-year fixed effects absorb nationwide shocks to each industry.

Detailed interpretation: The interaction coefficient compares the effect of the same industry-level delicensing shock across states with different labor regulations. Since higher labor regulation means more pro-worker, a negative β means delicensing works better in pro-employer states.

5.3 Instrumented labor regulation specification

The paper instruments labor regulation using historical and political variables, including interactions of post-1977 indicators with pre-1977 unionization and land-tenure patterns in British India.

Detailed interpretation: The instrumented specification addresses the concern that labor regulation is endogenous to state industrial performance. The point is not that the instruments are perfect, but that the main interaction survives a substantially different source of variation.

5.4 Trade and FDI robustness specification

To separate delicensing from trade and FDI reform, the paper estimates:

$$Y_{ist} = \alpha_{is} + \lambda_{st} + \mu_{it} + \beta(\text{Delicense}_{it} \times \text{LaborReg}_{st}) + \gamma(\text{Tariff}_{it} \times \text{LaborReg}_{st}) + \delta(\text{FDI}_{it} \times \text{LaborReg}_{st}) + \varepsilon_{ist}.$$

Detailed interpretation: The delicensing interaction remains stable after controlling for tariff and FDI interactions, indicating that the result is not an artifact of the broader 1991 external liberalization.

5.5 Economic interpretation of the sign

Because pro-worker labor laws raise firing or adjustment costs, firms in those states may be less able to expand, enter, reorganize, or reallocate labor after entry barriers are removed. Product-market liberalization therefore raises output more where labor-market constraints are weaker.

6 Identification and empirical design

6.1 What variation identifies the effect

The key variation is industry-level delicensing over time, interacted with state-level labor regulation. Delicensing is nationwide within an industry, but labor regulation varies by state. This creates a difference-in-differences-in-differences design.

6.2 Why nationwide industry reforms help

Because an industry is delicensed across all states at the same time, the reform is not chosen by individual states. This limits concerns that better-performing states selectively liberalized their own industries.

6.3 Fixed effects logic

The identification rests heavily on fixed effects:

- state-industry effects control for time-invariant state-industry comparative advantage;
- state-year effects control for state-level macroeconomic and policy shocks;
- industry-year effects control for nationwide industry-specific shocks.

6.4 Main identifying assumption

Conditional on these fixed effects and controls, changes in state-industries after delicensing would not have systematically differed by labor regulation for reasons unrelated to the reform. In other words, pro-worker and pro-employer states would not have had differential output changes in newly delicensed industries absent the delicensing shock.

6.5 Threats and responses

- **Omitted state capabilities:** addressed using state-year fixed effects and interactions with development expenditure and financial development.
- **Trade or FDI liberalization:** addressed by interacting tariff and FDI reforms with labor regulation.
- **Political environment:** addressed using interactions with party-majority variables.

- **Endogenous labor regulation:** addressed using pre-reform labor regulation and instrumented labor regulation.
- **Industry selection into delicensing:** partially addressed because timing is national and with industry-year fixed effects.

7 Tables and figures

7.1 Table 1: Sample Descriptive Statistics; Table 2: Average Effects of Delicensing on Industrial Performance in India: 1980–1997

Read: Table 1 documents the reform environment. By 1997, over 90 percent of three-digit industries and output are delicensed. Tariffs fall sharply after 1990, and the FDI reform measure rises after 1991. Manufacturing output, factories, employment, and fixed capital all rise over the sample.

Table 2 shows that delicensing increases the number of factories, but the average output effect is small and statistically insignificant. This is the central motivation for the heterogeneity analysis.

Economic message: The reform clearly changes the policy environment, but the average effect masks differences across states. The question becomes: why did some states benefit more than others?

Detailed interpretation: Table 1 tells us that delicensing is widespread and that other reforms occur at the same time. Table 2 shows why a simple average treatment effect is insufficient: entry rises, but average output does not reveal the unequal effects across institutional environments.

TABLE 1—SAMPLE DESCRIPTIVE STATISTICS

	1980	1985	1990	1997	1980–1997
<i>Delicensing</i>					
Percentage of three-digit industries delicensed	0	36.61	39.29	91.07	73.48
Percentage of real output delicensed	0	47.68	56.94	92.57	74.53
Percentage of employment delicensed	0	43.05	47.81	88.15	68.31
<i>Labor regulation</i>					
Labor regulation	−0.16 (1.04)	−0.05 (1.42)	0.13 (1.65)	0.13 (1.65)	0.04 (1.52)
<i>Trade liberalization</i>					
Tariff rate	119.19 (44.74)	142.31 (47.69)	132.53 (38.94)	47.58 (21.34)	117.62 (49.22)
<i>FDI reform</i>					
FDI reform	0	0	0	0.35 (0.39)	0.14 (0.30)
<i>Industrial performance</i>					
Mean log real output	11.47 (1.96)	11.88 (1.93)	12.31 (1.96)	12.68 (2.20)	12.13 (2.02)
Mean log number of factories	3.30 (1.34)	3.42 (1.29)	3.50 (1.30)	3.58 (1.36)	3.46 (1.32)
Mean log employment	7.22 (1.70)	7.37 (1.57)	7.46 (1.58)	7.55 (1.69)	7.43 (1.61)
Mean log real fixed capital	9.78 (2.12)	10.39 (2.03)	10.74 (2.12)	11.27 (2.46)	10.61 (2.20)

(a) Table 1: Sample Descriptive Statistics.

TABLE 2—AVERAGE EFFECTS OF DELICENSING ON INDUSTRIAL PERFORMANCE IN INDIA: 1980–1997

	(1) Log no. factories	(2) Log no. factories	(3) Log real output	(4) Log real output
Delicense	0.064*** (0.024)	0.064** (0.024)	0.032 (0.042)	0.031 (0.043)
Labor regulation		−0.062** (0.027)		−0.137*** (0.044)
Observations	18,324	18,324	18,324	18,324
R-squared	0.92	0.92	0.89	0.89
State-industry interactions	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

(b) Table 2: Average Effects of Delicensing on Industrial Performance.

7.2 Table 3: Delicensing, Labor Regulation, and Industrial Performance in India: 1980–1997; Table 4: Trade Liberalization, FDI Reform, Labor Regulation, and Industrial Performance in India: 1980–1997

Read: Table 3 is the main result. The coefficient on $Delicense \times LaborRegulation$ is negative and significant. Since higher labor regulation means more pro-worker, the result means that delicensing raises output more in pro-employer states. The result survives using 1980 labor regulation and instrumented labor regulation.

Table 4 shows that the main interaction is robust to controlling for trade liberalization and FDI reform interactions with labor regulation. The tariff and FDI interactions are not driving the main result.

Economic message: Product-market liberalization and labor-market flexibility are complements. Deregulating entry and capacity gives firms new opportunities, but pro-worker labor regulation limits firms’ ability to respond.

Detailed interpretation: Table 3 establishes the institutional heterogeneity result. Table 4 addresses the main historical confound: the early 1990s reforms also liberalized trade and FDI. The stability of the delicensing coefficient after adding these controls strengthens the interpretation that the effect is due to internal delicensing interacted with labor regulation.

TABLE 3—DELICENSING, LABOR REGULATION, AND INDUSTRIAL PERFORMANCE IN INDIA: 1980–1997

	(1) Log real output	(2) Log real output	(3) Log real output	(4) Log real output
Delicense $\times$ labor regulation	−0.054** (0.024)			−0.202** (0.078)
Delicense $\times$ 1980 labor regulation		−0.062* (0.035)		
Delicense $\times$ instrumented labor regulation			−0.068** (0.030)	
Observations	18,324	18,324	18,054	18,324
<i>R</i> -squared	0.92	0.92	0.92	0.92
State-industry interactions	Yes	Yes	Yes	Yes
State-year interactions	Yes	Yes	Yes	Yes
Industry-year interactions	Yes	Yes	Yes	Yes

(a) Table 3: Delicensing, Labor Regulation, and Industrial Performance.

TABLE 4—TRADE LIBERALIZATION, FDI REFORM, LABOR REGULATION, AND INDUSTRIAL PERFORMANCE IN INDIA: 1980–1997

	(1) Log real output	(2) Log real output	(3) Log real output	(4) Log real output
Delicense $\times$ labor regulation		−0.059** (0.024)		−0.059** (0.024)
Log tariff rate $\times$ labor regulation	0.009 (0.019)	0.008 (0.020)	0.009 (0.019)	0.008 (0.020)
FDI reform $\times$ labor regulation			−0.007 (0.038)	−0.010 (0.036)
Observations	17,783	17,783	17,783	17,783
<i>R</i> -squared	0.92	0.92	0.92	0.92
State-industry interactions	Yes	Yes	Yes	Yes

(b) Table 4: Trade Liberalization, FDI Reform, Labor Regulation, and Performance.

7.3 Table 5: Robustness to Interactions with State and State-Industry Characteristics

Read: Table 5 adds interactions between delicensing and state/state-industry characteristics, including development expenditure, financial development, industry productivity terciles, and political-party variables. The main delicensing–labor regulation interaction remains negative and statistically significant.

Economic message: The main effect is not simply picking up development spending, financial development, state politics, or pre-existing industry productivity differences.

Detailed interpretation: Table 5 is the broad robustness table. It is especially important because labor regulation may be correlated with many state-level policies and institutions. The stability of the main interaction suggests that labor regulation captures a distinct margin that shapes the gains from product-market liberalization.

TABLE 5—ROBUSTNESS TO INTERACTIONS WITH STATE AND STATE-INDUSTRY CHARACTERISTICS

	(1) Log real output	(2) Log real output	(3) Log real output
Delicense $\times$ labor regulation	−0.051** (0.024)	−0.064** (0.028)	−0.064*** (0.022)
Delicense $\times$ log development exp	0.188* (0.105)	−0.113 (0.101)	−0.118 (0.126)
Delicense $\times$ financial development	0.030** (0.014)	0.029* (0.017)	0.047** (0.023)
Delicense $\times$ top industry productivity tercile		0.472*** (0.032)	0.474*** (0.032)
Delicense $\times$ bottom industry productivity tercile		−0.521*** (0.033)	−0.523*** (0.033)
Delicense $\times$ Congress majority			−0.006 (0.005)
Delicense $\times$ hard-left majority			0.005 (0.020)
Delicense $\times$ regional majority			0.003 (0.006)
Delicense $\times$ janata majority			0.006 (0.019)

Table 5: Robustness to Interactions with State and State-Industry Characteristics. Single visual is set to width $0.6\backslash linewidth$ by rule.

7.4 Notes-created table: Empirical design map

Component	Role in design	Interpretation
Delicensing	Industry-year national reform	Removes central entry/capacity controls.
Labor regulation	State-year institutional environment	Captures pro-worker versus pro-employer adjustment costs.
Interaction	Main treatment intensity	Tests whether the same reform works differently across institutions.
State-industry fixed effects	Permanent comparative advantage	Controls for baseline location-industry differences.
State-year fixed effects	State-level shocks	Controls for statewide policies and macro shocks.
Industry-year fixed effects	National industry shocks	Controls for industry demand, technology, and nationwide policy timing.

Notes-created table: Empirical design map. This table is not an original table from the paper; it summarizes how the paper identifies heterogeneous effects of delicensing.

8 Short summary

- The paper studies India’s dismantling of the License Raj in registered manufacturing from 1980 to 1997.
- Delicensing was a national industry-level policy change; labor regulation varied across states.
- The average output effect of delicensing is small, but this hides strong heterogeneity.
- Delicensing raises output more in pro-employer states than in pro-worker states.
- The result is robust to state-year, industry-year, and state-industry fixed effects.
- It is also robust to controlling for trade liberalization, FDI reform, state development spending, financial development, political controls, and industry productivity.
- The main lesson is that product-market liberalization and labor-market institutions interact strongly.

9 Conclusion

9.1 Core conclusion

Aghion, Burgess, Redding, and Zilibotti show that liberalization has unequal effects across institutional environments. Dismantling the License Raj generated stronger manufacturing output growth in Indian states with pro-employer labor regulations and weaker growth in pro-worker states.

9.2 Economic intuition

Delicensing relaxes product-market constraints on entry, expansion, and capacity. But firms can exploit these opportunities only if they can adjust labor, reorganize production, and reallocate resources. Pro-worker labor regulations reduce this flexibility, so the same reform has smaller effects in those states.

9.3 Incremental contribution revisited

The paper's contribution is to demonstrate empirically that institutional complementarities matter for reform outcomes. The effect of deregulation cannot be evaluated independently from the surrounding labor-market environment. This insight helps explain why national reforms can produce uneven regional growth.

9.4 Policy implication

A policy maker should not expect product-market reform to have homogeneous effects. Complementary labor-market, financial, and state-capacity reforms may be needed if liberalization is to produce broad-based gains rather than widening disparities across regions.

9.5 Limitations

The design is not a randomized experiment. Labor regulation may still proxy for unobserved institutions, and the analysis focuses on registered manufacturing. The paper also measures output and formal-sector performance, not all welfare consequences for workers and informal firms.

9.6 Takeaway

The broad takeaway is that reform sequencing and institutional complementarities matter. Liberalization can raise output where firms have flexibility, but the same liberalization can leave more rigid regions behind.